

The fair-test repair lab

Teacher guide and worked answers

Plan and use a fair test of how card height affects shadow height.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

[_passsb/inputs/Build SOW 2026-2027.xlsx](#) | BUILD Weekly - Spring | C38

Plan and carry out a fair test about shadows or reflection.

Direct match to the named outcome. The lesson is a 40-minute exemplar, with optional greater depth; no accreditation completion is claimed.

Prior learning

Everyday objects can be described by what we observe.

Use a labelled diagram or evidence card to explain a choice.

Success evidence

Identify the intended change, measurement and controls.

Repair a plan that changes two factors.

Use repeated readings to make a supported conclusion.

Equipment

Low-power small LED source; fixed white screen; opaque cards 2, 4 and 6 cm high of same material and width; ruler; stands; printed plans and readings.

Preparation

Fix source-screen distance at 60 cm and card position at 30 cm from source. Keep cards upright, centred and parallel to screen.

Pre-cut three cards of the same width and material. Card height is the intended changed factor.

Pre-check that the shadow fits the screen and ruler. Provide constructed readings if the practical cannot run.

Practical care

Use a cool low-power LED below face height, never a laser, hot lamp or Sun.

Secure stands and card holders, keep walkways lit and clear. Adult checks the practical plan before use.

Use pre-cut cards with safe edges. Do not put a ruler or head in the beam near the source.

Ready to teach alternative

Analyse the flawed/repaiored plans and supplied constructed repeats. Pupils can complete a fair-test plan and conclusion without taking measurements.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Can this test answer its question?
2	Retrieval	2-5	Recall our shadow studio
3	I do	5-9	I spot two changed factors
4	I do	9-12	I repair the comparison
5	I do	Within stage	I measure and repeat consistently
6	We do	12-16	Make a test contract
7	We do	16-20	What should we do with 9?
8	Hinge	20-23	Which plan is fair for this question?
9	You do	23-25	Choose your investigation route
10	You do	Within stage	Use the evidence
11	You do	25-35	Make your own science record
12	You do	Within stage	Check and improve
13	Review	35-38	Lundy loop: improve our account
14	Exit	38-40	Three quick checks

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Can this test answer its question?

Invite pupils to act as method reviewers. The goal is a justified repair, not spotting a classmate's mistakes. Everyone uses the same supplied fictional team.

2 Retrieval Recall our shadow studio

Answers opaque card; yes with setup specified; centimetres. Recall that position matters before introducing the flawed plan.

3 I do I spot two changed factors

Think aloud: "The question is about card height. But I count two changes: height and distance. Distance can affect shadows, so I cannot say height alone caused any difference."

4 I do I repair the comparison

Think aloud: "I keep the same source and screen 60 cm apart. Each card stays 30 cm from the source. I also keep material, width and orientation the same. Now height is the intended difference."

5 I do I measure and repeat consistently

Shares I do time. Model zero at bottom and 8 cm at top. For repeats, check the same setup and read again; repositioning the card carefully between trials can check placement consistency. Do not silently switch to width or another ruler origin.

6 We do Make a test contract

Give pairs three headings: change / measure / keep same. Use plan cards A and B. Height is changed; shadow height measured; source-card and source-screen distances controlled, plus card material/width/orientation and source.

Reveal: Change card height; measure shadow height; keep card position the same.

7 We do What should we do with 9?

Use the shared reading table. 4 cm card has 8, 8 and 9 cm shadows. Do not erase 9 to make a perfect double pattern. Suggest checking ruler zero, edges, alignment and card position, then repeat. All three card-height groups still support taller card → taller shadow in this setup.

Reveal: Yes. Taller cards gave taller shadows here. Keep 9, check the setup and repeat if needed.

8 Hinge Which plan is fair for this question?

If A chosen, point back to distance effects. If C, explain that keeping the card height unchanged does not test its effect.

A. Change card height and source distance together. - Two changes mean we cannot separate their effects.

B. Change card height; keep the source, screen and card position fixed. - This isolates the intended change, with other relevant properties also controlled.

C. Keep card height the same and only change its colour. - That tests a different question, not the effect of height.

9 You do Choose your investigation route

Give one route only, selected with the pupil. The shared evidence page supports every route. Practical option after adult plan check: test each card at the fixed holder position and take three readings, using the same shadow-height rule. Keep actual values separate from the constructed table. Paper reviewers use the plans and readings to complete the full objective.

10 You do Use the evidence

Shares the 12-minute independent stage. Practical option after adult plan check: test each card at the fixed holder position and take three readings, using the same shadow-height rule. Keep actual values separate from the constructed table. Paper reviewers use the plans and readings to complete the full objective.

11 You do Make your own science record

Allow ten minutes of independent response after the two-minute setup. Practical option after adult plan check: test each card at the fixed holder position and take three readings, using the same shadow-height rule. Keep actual values separate from the constructed table. Paper reviewers use the plans and readings to complete the full objective.

12 You do Check and improve

Shares the independent stage. Ask the pupil to point to the clue supporting their answer; accept a spoken correction or changed choice.

13 Review Lundy loop: improve our account

Give a private or paired option. Adopt one scientifically sound pupil revision in the class explanation and name the change. No public ranking.

Reveal: A clear question, one intended change, controlled comparisons and honest repeated readings.

14 Exit Three quick checks

Collect brief individual answers. Secure core requires the main idea and a relevant reason. Revisit the relevant worked diagram next lesson if the explanation reverses cause and effect.

Reveal: See the teacher answer guide for the three checks.

Answers and assessment

Retrieval 1-3

1 opaque card. 2 yes, card position can affect size. 3 centimetres (cm).

We do W1-W2

W1 change card height, measure shadow height, control card position (30 cm) and source-screen distance (60 cm), with other relevant properties fixed. W2 retain 9, check method and repeat if useful; overall taller-card pattern remains supported.

Hinge

Second option correct. Also keep material, width, orientation and source the same. First option changes two factors; third tests colour instead of height.

S1-S5

S1 card height. S2 shadow height. S3 30 cm. S4 taller. S5 keep it, check the setup and repeat rather than erasing it.

T1-T5

T1 how does card height affect shadow height? T2 height; shadow height; any two relevant controls. T3 fix source-screen at 60 cm and card at 30 cm, use same material/width/orientation, change height, measure same top-to-bottom dimension, repeat three times and record all readings. T4 e.g. 2 cm card gave 4 cm shadows, 6 cm card gave 12 cm shadows; taller cards made taller shadows here. T5 retain it, check ruler zero/edges/position and repeat if needed.

X1-X4

X1 the 4 cm card includes a 9 cm reading, not exactly double. X2 shadow size also depends on source/card/screen geometry and source size. X3 in this setup, taller cards gave taller shadows, around twice card height. X4 check card position or ruler alignment/edge convention; take another reading for the 4 cm card after the check, retaining the original.

Exit E1-E3

E1 card height. E2 e.g. card-source distance, source-screen distance, card material/width/orientation or source. E3 no; keep honest readings and investigate differences.

Optional practical

Accept actual observations, including imperfect proportionality. Do not replace measurements with constructed sample values. Assess the fair comparison and evidence-based conclusion.

R1-R2

Responses vary. Credit a science-related revision or relevant next question. A pass or private response is acceptable; preferences have no fixed right answer.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 7; We do 8; hinge 3; You do 12; review 3; exit 2. Zero-minute slides share the stage time.

BUILD follows the user's Year 3 science anchor with age-respectful materials for secondary pupils. Routes are adjustable access choices, not fixed groups or diagnoses.

Every core task can run from the printed evidence. Do not present constructed teaching evidence as observations pupils made. A pupil's spoken reasoning carries the same scientific value as writing.

Do not require averages for the BUILD core. Pupils can compare repeated values, notice a range of 8-9 cm and write "around" or "about".

The ideal point-source geometry with source-screen distance twice source-card distance predicts doubled shadow height. The constructed 9 cm reading deliberately models measurement variation; it is not a reason to manufacture perfect data.

HTML We do: pupils choose a prediction, test it against the user-controlled SVG model or supplied evidence, then explain or revise. The shared paper cases and W1/W2 prompts provide the equivalent classroom discussion without a device. Any animation is a teaching model, not filmed experimental evidence.

Access and responsive teaching

Supported

Choose one route: short choices, word bank and pointing or adult-scribed explanation; keep the same science goal.

Standard

Use the evidence provided to record a decision and explain the science in short sentences.

Stretch

Apply the same core idea to a changed setup, identify a limitation and justify a better decision.

Regulation

Preview the sequence and equipment. Offer seated observation, a quiet paper route, partner rehearsal and a pass. Never require public performance or a darkened room.

Misconceptions to check

Any two different results show what caused the change.

If several factors change, their effects cannot be separated by that comparison.

Check: Ask what changed in the flawed plan.

A different repeat should be deleted.

Keep the reading, check the method and repeat if useful; do not remove it just to fit a pattern.

Check: Ask what to do with the 9 cm reading.

Fair means everybody must get exactly the same number.

Fair concerns the comparison; measurements can vary even with a careful method.

Check: Ask why repeated measurements are useful.

Word	Meaning
fair test	A comparison changing one chosen factor while keeping relevant others the same.
repeat	Do a measurement again under the same conditions.
evidence	Information used to support an explanation or decision.

Scientific references

Department for Education: science programmes of study

Checked 8 September 2026. Year 3 light outcomes anchor the BUILD content; ray explanations are carefully modelled as an accessible extension.